Build: from a clean machine to an installed application

The project targets **Linux x86-64**, Python **>=3.12**. The delivery formats are described in docs/INSTALL.md and docs/BINARY.md.

Pinned versions

Tool	Version	Source of truth
Python	3.12 and 3.13 (the CI matrix)	`pyproject.toml` (`requires-python`), `ci.yml`
PySide6	6.8.1	`pyproject.toml`
dpkt	1.9.8	`pyproject.toml`
PyYAML	6.0.2	`pyproject.toml`
dev tools	pytest 9.0.3, ruff 0.16.10, mypy 2.4.0, pytest-cov 7.1.0, pip-audit 2.10.1	`[project.optional-dependencies].dev`
web stack	fastapi, uvicorn, jinja2, python-multipart	`[project.optional-dependencies].web`

The `web` and `dev` requirements are given as ranges: for a reproducible release environment pin them with `pip freeze` when building the artefact — a deliberate trade-off (see docs/audit/portability-audit-2026-10-29.md).

Level A — source code

```
make doctor          # python, dpkt/PyYAML, PySide6, libEGL, QT_QPA_PLATFORM, ruff, LICENSE
make install-dev     # pip install -e ".[dev,web]"
make lint            # ruff
make test-core       # 446 tests without Qt: protocol, capture, hypothesis (~15 s)
make test-web        # the web interface
make docs-check
```

Level A works without a graphical subsystem: the core does not depend on Qt.

Level B — artefacts

```
make wheel           # dist/*.whl + sdist
make package         # install the wheel into a clean venv and run madrigal-lab --help
```

Level C — the interface

```
make test-ui         # QT_QPA_PLATFORM=offscreen, needs PySide6 and libEGL
```

On a server without a display the interface runs with `QT_QPA_PLATFORM=offscreen`.

Level D — the full test set

```
make test            # all tests/ directories (as in CI)
```

Level E — installation on a clean machine

```
python -m venv /tmp/venv && . /tmp/venv/bin/activate
pip install dist/*.whl
QT_QPA_PLATFORM=offscreen madrigal-lab --help
```

The delivery formats (a binary, a `.deb`-like layout, a desktop file, an installer) are described in docs/INSTALL.md and `packaging/`.

Typical errors

Error	Cause	Solution
`ImportError: libEGL.so.1`	the system Qt libraries are missing	install `libegl1 libgl1 libxkbcommon0`
`qt.qpa.plugin: could not load the Qt platform plugin "xcb"`	no display	`QT_QPA_PLATFORM=offscreen`
The UI tests fail on a runner	no PySide6 or no locale	the CI matrix installs the Qt libraries; locally `make install`
`ModuleNotFoundError: src`	run without installing the package	`make install-dev` or `PYTHONPATH=src`
`pytest` finds no tests	the venv is not activated	activate `$(VENV)`

Proof

`ci.yml` (tests, interface, code quality, dependency audit) and `portability.yml` (packaging, an installation check, documentation). The claim matrix — docs/verification/matrix.md.